log in software is base 2 but in mathematics is base 10 why?

Great question! The difference in logarithm bases between **software/computer science** and **mathematics** comes down to **how numbers are represented and manipulated** in each field.

1️⃣ Base-10 (Common Logarithm in Mathematics)

- In traditional mathematics, the logarithm **$\log_{10}(x)$** is commonly used because our numerical system is **decimal (base 10)**.
- It's natural for humans since we count using **digits 0–9**, making calculations intuitive.
- Logarithms in base 10 are widely used in **engineering, physics, finance**, and real-world measurements.

2️⃣ Base-2 (Logarithm in Software & Computer Science)

- Computers use **binary (base-2)** representation, meaning all data is stored as **0s and 1s**.
- In computing, many fundamental operations involve **powers of 2** (e.g., memory sizes, algorithms, data structures).
- When analyzing algorithms, we often use **$\log_2(x)$** because it directly relates to how efficiently computers process information.
  - Example: **Binary search** has time complexity **$O(\log_2(n))$** because each step halves the remaining search space.

Why the Difference?

Computers work with **binary**, so **base-2 logarithms** make sense for analyzing efficiency (number of bits needed, complexity of operations). Humans use **decimal**, so **base-10 logarithms** are more intuitive for everyday calculations.

In some cases, **natural logarithm ($\log_e$ or ln)** is preferred in advanced mathematics and science because it relates to **continuous growth processes**, but in computing, binary dominates.